

Jian Xiao

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EDUCATION

Washington University in St. Louis

Aug. 2025 – Present

Ph.D. in Computer Science Engineering

St. Louis, MO

Key Courses: Recent Advances in Computer Security and Privacy, System Security, Introduction to Computer Security

Duke University

Aug. 2022 – May 2024

Master of Engineering in Computer Engineering, GPA:3.91/4.0

Durham, NC

Key Courses: Algorithms, Computer System, Robust Server, Reliable System, Advanced Computer Network

Beijing Jiaotong University

Sep. 2017 – June 2021

Bachelor of Engineering in Software Engineering, GPA:3.9/4.0 (top 3% out of 172)

Beijing, China

Key Courses: Calculus, Discrete Mathematics, Probability Theory and Mathematical Statistics, Geometry and Algebra, Computer Network, Operating System

RESEARCH EXPERIENCE

Three-Layer Federated Learning with Momentum to Solve Non-IID

Dec. 2023 – Aug 2025

Research Assistant; Advisor: Dr. Enmao Diao and Prof. Jie Ding

University of Minnesota (ColAI)

- Proposed a three-layer structure **Federated Learning** algorithm(local, fed, global) with momentum to address **non-IID** issues, managing variance at different levels—local for within-client iterations, fed level for coordination across clients, and global for coordination across iterations. This innovative structure can be simplified to the basic FedAvg algorithm through hyperparameter adjustments.
- Enhanced the FLPipe code to integrate **Ray** for federated learning. Deployed Ray on **AWS** and modularized the federated learning logic into client, controller, and server files. The controller manages the server and its local database, enabling experiment resumption and testing through local parameter saving and restoration.
- Utilized SGD and CosineAnnealingLR with full gradient descent in a CIFAR-10 CNN setup with non-IID data. By adjusting only the local learning rate and momentum, a maximum accuracy of **79.6000** is achieved, representing an extreme case. Further tuning of the fed level's parameters increases accuracy to around **82.0000**, while the IID result is approximately 84.0000. In typical scenarios, the fed lever significantly improves non-IID performance, narrowing the gap to IID accuracy.
- Implemented a code resumption mechanism, enabling complete replication of results under round-robin sampling strategy after interruptions, ensuring consistent experimental outcomes.
- Designed and implemented a robust benchmarking framework for evaluating advanced federated learning algorithms, encompassing a diverse set of neural network architectures, optimizers, datasets, and hyperparameter configurations to provide a comprehensive assessment of performance and scalability.
- Currently working on porting clientfedDyn, fedGen, FedProx, FedNova, and SCAFFOLD into the proposed framework to expand its functionality and adaptability.

DXF Drawings AutoGrader

April. 2024 – Aug 2025

Research Assistant; Advisor: Prof. Rabih Younes

Duke University

- Designed and deployed a scalable CAD grader using React and Flask to automate the assessment of DXF files, ensuring efficiency and fairness in grading. Utilized **Nginx**, **Gunicorn**, and systemctl for deployment and Locust for performance testing on a Linux server.
- Leveraged the ezdxflib library for automated detection of geometric entities and error types, including color, linewidth, scale, rotation, and hatching errors, ensuring modular and extensible back-end development.
- Exploring the use of **CNN** for image recognition and geometric feature extraction, alongside **GNN** for precise error detection and classification of entities, scaling, and rotation errors, with plans to enhance model accuracy and generalization through advanced data augmentation techniques.

Students Acceptance Optimizer

May. 2023 – Oct. 2023

Research Assistant; Advisor: Prof. Rabih Younes

Duke University

- Developed a student acceptance optimizer that can predict students' first-year GPA at Duke based on their application metrics such as GPA, undergraduate institution, GRE, TOEFL, etc., to provide references for the ECE admissions team.

- Implemented two-level cross-validation to select the optimal **LASSO regression model** which has the highest coefficient of determination score. Fine-tune the pre-trained **BERT model**, and based on the obtained sentiment analysis scores, label each letter of recommendation into three categories: positive, negative, and neutral.
- Developed a user-friendly GUI interface using **PyQt**. Utilized the **Nuitka** tool to package the Python project and generate executable .exe files or binary files, ensuring compatibility for both Windows and Mac OS platforms.

Improved Two-stream Network for Action Recognition

June 2020 – May 2021

Research Assistant; Advisor: Prof. Weiwei Xing

Beijing Jiaotong University

- Participated in developing a multi-stream model combining **LSTM** and **DenseNet** for action recognition. The model uses RGB video frames and optical flow, with a spatial stream for frame analysis, a temporal stream for flow processing, and an SVM fusion layer for final classification.
- Contributed to integrating CNN and LSTM in the spatial stream for capturing both spatial and temporal patterns from RGB video frames, enhancing the model's ability to recognize actions accurately.
- Achieved **91.25%** accuracy on the UCF-101 dataset using an improved two-stream network with DenseNet and LSTM,

Computer Vision-Based Railway Signal Light Distance Measurement

June 2020 – May 2021

Research Assistant; Advisor: Prof. Weiwei Xing

Beijing Jiaotong University

- Proposed distance measurement method to detect distance between train and signal light according to signal light position and railway track line expression, which can help drivers make better decisions and avoid risks
- For the signal lamp component, utilized different datasets to handle single and multiple signal lamps more effectively. Applied **OpenCV** for the recognition method based on Hough transform and left-right track line screening for straight tracks, and used a recognition method based on perspective transform and fitting for curved tracks.
- Experimented under multiple weather conditions and different distances. Achieved error rate as low as 3%.
- Awarded as excellent thesis recommendation (**Top 10%**) for senior thesis.

COURSE PROJECTS

HTTP Caching Proxy Server | C++, Socket Programming

Jan. 2023 – March. 2023

- Developed an HTTP Caching Proxy Server that can handle GET, POST, CONNECT requests using **C++**.
- Used **multi-thread** to handle requests from different clients. Used **Read-Write Lock** to manage logs. Used **Socket Programming** to handle packets. Used **RAII/smart pointer** to ensure strong exception guarantee.
- Improved the performance of Proxy Server by caching responses with **LRU** cache management policy. Validation and expiration rules(mainly including Etag, Last-modified to implement re-validation, logic processes of max-age, no-store, no-cache, must-revalidate in Cache-control field) in cache policy came from RFC7324.

TECHNICAL SKILLS

Languages: C/C++, Python(Pytorch, Scikit-learn, Pandas, Numpy), Java, SQL, JavaScript, HTML/CSS

Developer Tools: Git, Linux, Vim, Docker, Kubernetes, Visual Studio, PyCharm, AWS, Ray, CUDA

AWARDS

University-Level Outstanding Student Award	Dec 2018, 2019, 2020
First-class Outstanding Academic Performance Scholarship (3%)	Dec 2020
Second-class Outstanding Academic Performance Scholarship (10%)	Dec 2018, 2019
Wntime Scholarship of Beijing Jiaotong University (2/171)	Dec 2020
School-level Project Approval, National College Students Innovation and Entrepreneurship Competition	May 2019
Municipal Project Approval, National College Students Innovation and Entrepreneurship Competition	Sep 2020